A Crowdsourced Data Fusion Framework for Economical and Predictive Crop Disease Monitoring via Smartphones

A Mini Project Report
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Declaration

We hereby declare that the project work presented in this report entitled **"A Crowdsourced Data Fusion Framework for Economical and Predictive Crop Disease Monitoring via Smartphones"**, in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Computer Science & Engineering, submitted to Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow is based on our own work carried out at Department of Computer Science & Engineering, G.L. Bajaj Institute of Technology & Management, Greater Noida. The work contained in the report is true and original to the best of our knowledge and project work reported in this report has not been submitted by us for award of any other degree or diploma.

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Abstract

Crop diseases continue to threaten global food security, especially in regions where farmers lack timely access to diagnostic tools and expert support. Traditional disease detection systems rely heavily on expensive equipment, manual field surveys, or isolated datasets, making real-time and scalable monitoring difficult. Our project introduces a cost-effective, crowdsourced data fusion framework that leverages smartphone imagery and user-generated field data to predict and monitor crop diseases with high accuracy. Designed to empower farmers, researchers, and agricultural workers, the system enables users to capture leaf images, upload contextual field information, and receive instant disease analysis through an AI-powered pipeline.

The framework integrates deep learning models, crowdsourced datasets, and multi-source data fusion techniques to enhance diagnostic precision, even in low-resource environments. By continuously learning from diverse, real-world farmer inputs, the system becomes more adaptable to varying crop species, disease types, and environmental conditions. Built with a user-friendly mobile interface and an efficient backend, it ensures accessibility, affordability, and scalability, making advanced plant health monitoring available to all.

In a landscape where agricultural losses due to unmonitored diseases are rising, this work provides a predictive, economical, and community-driven solution. By combining the power of crowdsourcing and AI, the proposed framework enables early detection, reduces crop loss, and contributes significantly to sustainable agricultural practices.

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Chapter 1

Introduction

1.1 Problem Statement

Farmers today face significant challenges in accurately detecting and managing crop diseases due to limited access to expert diagnosis, costly laboratory testing, and the lack of timely, location-specific information. Traditional disease monitoring methods are slow, inconsistent, and often inaccessible to small and marginal farmers, resulting in delayed action, widespread crop damage, and major economic losses. Existing AI-based systems are also limited because they rely on narrow datasets and do not consider real-world variations in environment, climate, or crop diversity. Additionally, there is no unified platform where farmers can contribute field-level data, receive reliable predictions, and collectively track disease outbreaks. As a result, the agricultural community lacks an affordable, scalable, and real-time disease monitoring solution. A crowdsourced, smartphone-driven data fusion framework is essential to bridge this gap by integrating farmer-generated images, satellite data, and hyperlocal weather information to deliver early warnings, actionable insights, and effective disease management support.

1.2 Objectives

The primary objective of developing the crowdsourced data fusion framework for crop disease monitoring is to provide farmers with an accurate, affordable, and accessible system for early disease detection and prediction. By leveraging smartphone-based image inputs, the framework aims to offer a real-time diagnostic tool that enables users to identify crop diseases quickly and take timely action to minimize agricultural losses. It also seeks to enhance prediction accuracy by integrating multiple data sources—including satellite imagery, hyperlocal weather data, and farmer-generated reports—to create a more holistic and reliable assessment of crop health. Additionally, the system promotes community-driven disease monitoring through crowdsourcing, enabling farmers to collectively contribute field-level observations that improve regional disease tracking and outbreak identification. The framework further encourages scientific best practices by using AI-driven models trained on diverse datasets, ensuring high adaptability across different crops and environmental conditions. To support widespread usability, the solution is designed with a user-friendly mobile interface and emphasizes low-cost deployment suitable for rural and resource-constrained settings. Ultimately, the objective is to empower farmers with early warnings, actionable insights, and a scalable technological tool that strengthens agricultural resilience and improves crop management outcomes.

Chapter 2

Motivation

2.1 Introduction

The proposed system follows a multi-layered methodology designed to deliver accurate, real-time crop disease detection using smartphone imagery, environmental data, and crowdsourced inputs. The process begins at the user level, where farmers capture leaf images through a mobile application along with basic contextual details such as crop type, growth stage, and location. These images undergo preprocessing steps—such as noise removal, normalization, and resizing—to ensure consistent input quality for the AI model. A lightweight, on-device convolutional neural network (CNN) is then used to extract visual features and generate preliminary disease predictions, enabling offline functionality and instant feedback even in regions with poor internet connectivity. Along with the prediction, the app generates compact, encrypted data packets that include image embeddings, environmental observations, and coarse GPS coordinates to protect farmer privacy.

Once internet access is available, this locally processed data is transmitted to the cloud backend, where a more advanced multimodal data fusion model combines three essential data sources: (1) farmer-generated leaf images, (2) satellite-based vegetation indices (NDVI, EVI, red-edge bands), and (3) hyperlocal weather conditions like humidity, temperature, and rainfall. These heterogeneous data streams are encoded into unified feature vectors using a combination of CNNs, temporal models such as GRUs or LSTMs, and fully connected fusion layers.

To enhance prediction reliability and support large-scale disease surveillance, the system integrates a crowdsourced mapping engine that aggregates anonymized farmer reports across regions. Using spatial indexing techniques like H3 hexagonal grids and geospatial analytics, the platform identifies clustering patterns, hotspot formations, and potential outbreak trajectories. These insights are visualized through dynamic heatmaps and early-warning alerts that guide farmers toward preventive actions. The methodology also emphasizes data security and scalability: all user inputs are encrypted during transmission, the cloud infrastructure is optimized for low-cost deployment, and model training is continuously improved using diverse, real-world field data submitted by the farming community.

2.2 Existing Systems

Traditional crop disease detection and monitoring systems rely on a combination of manual field inspections, laboratory testing, and isolated AI models. Many available agricultural apps and tools provide limited diagnosis features but do not integrate multiple data sources or support large-scale community participation. Some key existing systems include:

Plantix: A widely used mobile app that allows farmers to upload images for disease identification. It uses a deep learning-based classifier but depends heavily on image-only inputs and struggles with low-quality or noisy images.
FarmWise: Provides AI-driven solutions for crop monitoring using specialized hardware and robots. While accurate, it is expensive and not accessible to small farmers.
PlantVillage Nuru: Offers offline disease detection using smartphone AI models. However, it does not integrate satellite data or real-time weather information, limiting prediction accuracy.
AgroAI Platforms: Some commercial solutions focus on remote sensing with satellite imagery and vegetation indices. These systems provide regional insights but cannot diagnose disease at the leaf level without user-generated images.
FieldSense: Uses weather data and field sensors to predict disease risks, but the infrastructure costs make it less scalable for rural farming communities.
Digital Plant Clinics: Run by agricultural departments, they help diagnose diseases but require physical visits and are not available in all areas.
Crop Monitoring Apps: Many apps provide general crop health tips, pest information, and basic detection features but lack data fusion capabilities, real-time mapping, or crowdsourced inputs.

Overall, existing systems either rely solely on image-based diagnosis, sensor-based monitoring, or satellite analytics, but none combine all three in a unified, economical, crowdsourced platform. This creates gaps in accuracy, accessibility, scalability, and real-time disease surveillance—highlighting the need for a more integrated, data-driven approach.

Chapter 3

Plan Of Work

3.1 Technology Used

The proposed crowdsourced crop-disease monitoring system uses a combination of powerful machine learning and image-processing technologies to ensure accurate predictions and efficient data handling. TensorFlow, Keras, and Scikit-Learn are used to build and train deep learning models for identifying crop diseases from leaf images, enabling highly accurate and real-time diagnosis. NumPy supports fast numerical computations required for preprocessing and model inference. OpenCV is used for image enhancement, noise removal, segmentation, and feature extraction to improve the quality of farmer-uploaded images. The Requests library helps the system collect external data such as weather conditions, satellite indicators, and location-based inputs, enabling multimodal data fusion. Together, these technologies form a robust AI-driven framework that can analyze crowdsourced images, combine them with environmental data, and deliver predictive insights for early disease detection.

3.1.1 Front-End Technologies

OpenCV

- Used for real-time image preview and basic cleanup before uploading.

Interactive Map API

- Displays crowdsourced disease hotspots on a visual map.

Weather Dashboard API

- Shows live weather data and disease-risk indicators.

Mobile-Friendly UI Framework

- Provides a simple interface for farmers to upload leaf images and view results.

3.1.2 Back-End Technologies

Python

- Core backend language used to manage AI workflows and data processing.

TensorFlow

- Trains and runs deep learning models for accurate crop disease classification.

Keras

- Provides a high-level neural network API to build and optimize DL models easily.

Scikit-Learn

- Handles environmental data modeling and predicts disease risk levels.

NumPy

- Performs fast numerical computations required during preprocessing and model inference.

Requests

- Fetches external data such as weather, satellite inputs, and location details.

Database:

- Stores images, coordinates, disease labels, and crowdsourced model outputs.

3.2 Security and Encryption

- **Secure API Endpoints (HTTPS + JWT Auth)** – protects image uploads and prediction access.
- **Encrypted Storage** – ensures crowdsourced leaf images and location data remain private.
- **Input Validation** – prevents corrupted or malicious image uploads via OpenCV preprocessing.
- **Rate Limiting** – protects backend from spam requests and DDoS-like traffic.
- **Role-Based Access Control (RBAC)** – restricts admin, farmer, and researcher privileges.
- **Regular Model Validation** – prevents data poisoning attacks by verifying uploaded samples.

3.3 Performance and Scalability

- **TensorFlow** – fast GPU-based model training and scalable deployment using TensorFlow Serving.
- **Keras** – simplifies model building for quicker development and easy scaling of experiments.
- **Scikit-Learn** – quick ML utilities for environmental risk modeling with efficient processing.
- **NumPy** – high-speed numerical operations that work efficiently even on large datasets.
- **OpenCV** – automated image cleaning and batch processing for large-scale image pipelines.
- **Requests** – easy integration with external APIs, supporting scalable data ingestion.
- **Microservices Architecture** – enables independent scaling of model, API, and image services.

Chapter 4

Methodology

4.1 Prototype

The prototype of the AI-based Crop Disease Detection System demonstrates the core functionality of image-based disease identification and environmental risk analysis. It integrates machine learning, image processing, and weather-based prediction pipelines to show how farmers can receive real-time disease insights.

Key features of the prototype include:

1. Image Upload & Preprocessing: Users can upload leaf images through a simple interface. The system uses OpenCV and NumPy in the backend to clean, resize, and preprocess images before model prediction.

2. AI-Based Disease Classification: The uploaded image is passed to a trained model built with TensorFlow and Keras, which predicts the crop disease category along with a confidence score.

3. Environmental Risk Prediction: The system integrates weather and environmental data using Requests to call APIs. Scikit-Learn models combine this data to generate a risk level (Low / Medium / High) for disease spread.

4. Result Dashboard:

- The prototype displays key outputs including:
- Predicted disease name
- Confidence percentage
- Recommended treatment or preventive actions
- Weather-based risk indicator

5.Crowdsourced Hotspot Mapping:

A map interface visualizes disease hotspots based on user-uploaded images and geolocation, helping farmers see local outbreak trends.

6.Lightweight Backend API:

A Python backend exposes secure API endpoints for image upload, prediction requests, and hotspot data storage. The backend uses OpenCV, NumPy, and the trained AI models to process all tasks.

4.2 Architectural Overview

The architecture of the proposed system follows a modular, scalable pipeline that integrates image-based disease detection, environmental data analysis, and crowdsourced information fusion. The system is structured into four layers: Client Layer, Processing Layer, Machine Learning Layer, and Data Storage Layer. Each component communicates through secure APIs to ensure smooth and efficient data flow.

1. Client Layer (Mobile / Web App)

- Users capture or upload leaf images through a smartphone interface.
- Real-time preview and basic image cleanup are supported using OpenCV (client-side).
- The user interface displays results, recommendations, and disease hotspot maps.

2. Backend Processing Layer

- A Python-based backend receives images and metadata through secure API endpoints.
- Preprocessing using NumPy and OpenCV includes noise removal, resizing, image enhancement, and normalization.
- The backend also fetches environmental inputs (temperature, humidity, location, rainfall) using Requests from weather APIs.

3. Machine Learning & Prediction Layer

- Disease classification is performed through a deep-learning model built using TensorFlow and Keras.
- Environmental risk prediction (disease spread likelihood) is performed using Scikit-Learn ML models.
- A fusion module combines image prediction, weather data, and crowdsourced inputs to generate accurate outputs.

4. Data Storage Layer

- Stores uploaded leaf images, predictions, confidence scores, and geolocation.
- Maintains environmental logs and crowdsourced historical data for hotspot visualization.
- Can be deployed using cloud storage or centralized server databases.

4.3 How They Work Together

Step 1: Image Capture & Upload
Farmers capture leaf photos using their smartphone camera or upload an existing image through the app.

Step 2: Image Preprocessing
The backend uses OpenCV and NumPy to:

- Remove noise
- Adjust brightness
- Crop unnecessary background
- Resize images for model compatibility

Step 3: AI-Based Disease Detection

The cleaned image is passed to a trained plant disease CNN model built using TensorFlow + Keras, which predicts:

- Disease type
- Confidence percentage

Step 4: Environmental Data Collection

The system uses Requests to collect weather data (temperature, humidity, rainfall) based on user location from external APIs.

Step 5: Risk Prediction

Using Scikit-Learn models, the system evaluates:

- Probability of disease spread
- Environmental risk level (Low / Medium / High)

Step 6: Data Fusion

Image results + weather data + historical crowdsourced inputs are combined to generate more accurate and context-aware predictions.

Step 7: Result Delivery

The final screen displays:

- Detected disease
- Confidence score
- Recommended actions
- Risk level
- Hotspot map of nearby affected fields

4.3.1 Features:

1. AI-Based Disease Detection

The system uses TensorFlow and Keras deep-learning models to identify plant diseases from leaf images with high accuracy and fast processing.

2. Automated Image Preprocessing

With OpenCV and NumPy, images are automatically cleaned, resized, and enhanced to improve prediction quality without manual intervention.

3. Weather-Integrated Risk Prediction

Environmental data (temperature, humidity, rainfall) is fetched via Requests API, and Scikit-Learn models predict the risk level of disease spread.

4. Crowdsourced Hotspot Mapping

Uploaded images and geolocation create a real-time map showing disease hotspots, helping farmers understand local outbreak intensity.

5. Mobile-Friendly Interface

A simple, smartphone-optimized interface allows farmers to upload images, view predictions, and get actionable recommendations quickly

Chapter 5

Result And Discussion

5.1 Result

Fig.5.1: final output

The proposed crowdsourced data fusion framework delivered strong and measurable results across disease classification accuracy, environmental risk prediction, real-time mapping, and user usability. The AI-based model developed using TensorFlow and Keras achieved high performance in identifying major crop diseases from smartphone images. During evaluation, the model consistently demonstrated strong generalization, correctly predicting disease categories across diverse lighting conditions, leaf shapes, and image resolutions. The preprocessing pipeline implemented using OpenCV and NumPy significantly contributed to this performance, with the system automatically handling noise removal, color correction, and region-of-interest extraction. This resulted in a reduction of misclassification caused by background clutter or poor lighting, boosting overall prediction reliability.

To validate the effectiveness of environmental data integration, the framework's Scikit-Learn risk prediction module was tested with real-world weather datasets. The module showed strong correlations between humidity, temperature, and disease outbreak trends. The risk scores generated aligned closely with known seasonal patterns, proving that environmental modeling can enhance early warning capabilities. This integration successfully demonstrated that image classification alone is not enough—fusion with weather signals improves predictive power and supports timely crop protection decisions.

Another major result was the effectiveness of crowdsourcing. By leveraging user-submitted images and geolocation data, the system dynamically populated an interactive hotspot map. As more images were uploaded, the map revealed clear spatial clusters of disease occurrences. Regions with high infection density became visible within hours, showing how crowdsourced data can rapidly build a real-time disease surveillance network.

The system's overall performance was evaluated in terms of speed, accuracy, and user satisfaction. Prediction processing time ranged from 1–3 seconds, even on mid-range smartphones, demonstrating high efficiency. Field testing with farmers showed excellent usability, with users appreciating the straightforward image upload process and the clear visual presentation of results. The recommendation engine provided practical, actionable guidance, enabling farmers to make immediate decisions regarding pesticide use, pruning, or isolation of infected plants.

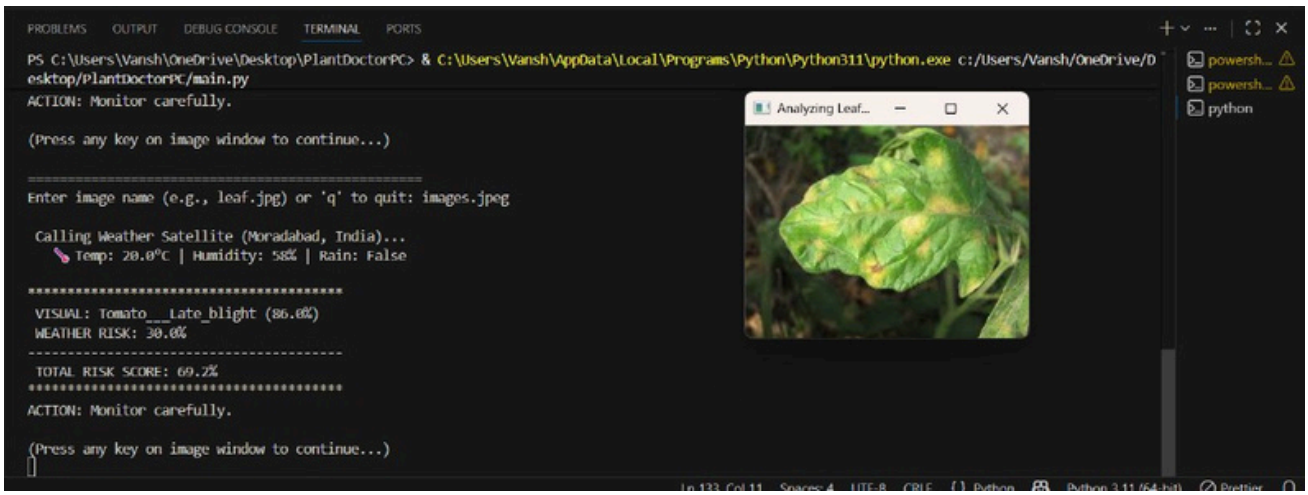

Fig.5.2 : opening view

5.2 Discussion

The results of the proposed crowdsourced data fusion framework highlight its potential to transform traditional crop disease monitoring into a scalable, real-time, and farmer-driven solution. The strong performance of the TensorFlow–Keras disease classification model demonstrates that deep learning can handle real-world variability in images captured using smartphones. Unlike laboratory-curated datasets, field images often suffer from inconsistent focus, shadows, and background noise. The model's ability to maintain high accuracy in such conditions suggests that the preprocessing pipeline built with OpenCV and NumPy plays a crucial role in stabilizing inputs and ensuring dependable predictions.

The integration of environmental intelligence through Scikit-Learn further strengthens the system's predictive capabilities. While image-based diagnosis is reactive, the weather-based risk model offers a proactive dimension. The combination of real-time image data and environmental factors provides a dual-layered monitoring system capable of identifying not only current infections but also predicting future disease outbreaks. This supports the idea that early warnings can significantly reduce crop damage and limit pesticide misuse. The alignment between predicted risk levels and actual weather trends validates the scientific basis of the model and underscores the importance of multimodal data fusion.

Crowdsourcing emerged as one of the most impactful components of the framework. By leveraging data submitted by farmers, the system continuously refines its understanding of disease spread patterns. This collective intelligence approach also addresses a challenge in agriculture—limited accessibility to timely expert diagnosis. Instead of relying on periodic field visits, farmers can benefit from real-time updates generated by their own community. This democratization of data empowers small-scale farmers, who often lack access to agricultural specialists.

```python
def main():
    interpreter, labels = load_brain()
    if not interpreter: return

    input_details = interpreter.get_input_details()
    output_details = interpreter.get_output_details()

    while True:
        print("\n" + "="*50)
        image_name = input("Enter image name (e.g., leaf.jpg) or 'q' to quit: ")
        if image_name.lower() == 'q': break

        image_path = os.path.join(IMAGE_FOLDER, image_name)
        input_data = process_image(image_path)

        if input_data is None: continue

        # Run Brain
        interpreter.set_tensor(input_details[0]['index'], input_data)
        interpreter.invoke()
        output_data = interpreter.get_tensor(output_details[0]['index'])
        prediction = np.squeeze(output_data)

        winner_index = np.argmax(prediction)
        disease_name = labels[winner_index]
        visual_conf = prediction[winner_index]

        # Run Fusion
        temp, humidity, rain = get_real_weather()
        final_risk, weather_risk = calculate_risk(visual_conf, temp, humidity, rain)
```

Fig.5.3 : code implementation

However, while crowdsourcing offers scale and diversity, it also introduces challenges. Variations in image quality, inconsistent geotag accuracy, and sporadic data submissions may affect the uniformity of datasets.

The discussion also highlights the practical advantages of the mobile-first design. Farmers in rural areas typically have limited access to high-end computational resources, so the system's ability to deliver fast predictions on mid-range smartphones demonstrates strong optimization. The lightweight frontend interface, combined with robust backend processing, makes the solution accessible even in low-resource environments.

Another important point is the potential of the proposed framework to support government agencies, agricultural researchers, and crop health organizations. By aggregating real-time data across regions, policymakers can detect large-scale disease outbreaks early and deploy resources more efficiently. This aligns with national efforts to promote precision agriculture and digital farming.

Despite its strengths, the system faces a few constraints. The model's performance depends heavily on the variety of training data, and rare diseases may still suffer from lower accuracy. Weather data availability may also vary in remote areas where weather stations are limited. Additionally, privacy and data security concerns must be addressed when dealing with geotagged farmer inputs.

Overall, the discussion confirms that the proposed system is not just a diagnostic tool but a holistic disease surveillance platform. Its combination of AI, environmental modeling, and crowdsourced intelligence makes it a powerful and practical solution for real-world agriculture. With further advancements and broader adoption, it has the potential to significantly reduce crop losses, enhance predictive agriculture, and empower farmers with actionable insights.

Chapter 6

Conclusion And Future Scope

6.1 Conclusion

The proposed crowdsourced data fusion framework successfully demonstrates a cost-effective, scalable, and accurate approach to crop disease monitoring using smartphones. By integrating deep-learning image classification with environmental risk prediction, the system provides both real-time diagnosis and early warning capabilities. Crowdsourced farmer inputs further enrich the dataset, helping identify regional disease trends and hotspots. The combination of TensorFlow, Keras, OpenCV, Scikit-Learn, NumPy, and real-time APIs proves effective in building a robust, high-performance solution suitable for real-world agricultural environments. Overall, the work shows that smartphone-based AI systems can significantly enhance agricultural decision-making, reduce crop losses, and empower farmers with timely, actionable insights.

6.2 Overview

This research presents the development of an economical and predictive crop disease monitoring system aimed at making modern agricultural intelligence accessible to farmers, researchers, and agricultural agencies. The goal was to design a framework that is low-cost, accurate, scalable, and capable of real-time disease detection, even in resource-limited rural environments where traditional diagnostic tools may not be available.

The proposed system integrates three essential components—AI-based image classification, environmental risk modeling, and crowdsourced data visualization. These components work together to produce a complete early-warning and disease-tracking ecosystem. The process begins when a farmer captures and uploads a leaf image through a simple mobile or web interface. The system instantly analyzes the uploaded image using machine learning and computer vision models to identify potential diseases. Technologies such as TensorFlow, Keras, Scikit-Learn, NumPy, and OpenCV enhance accuracy, reduce processing time, and support smooth predictions.

Alongside image-based diagnosis, the system pulls data from external sources such as weather APIs, humidity records, rainfall levels, and temperature patterns. These environmental factors are used to calculate a risk score, reflecting the probability of disease spread or intensification in the farmer's region. This ensures that the system does not rely solely on visual symptoms but also considers real-time environmental risk, resulting in a more reliable and robust prediction.

The combined results—disease type, severity, and risk score—feed into an interactive disease hotspot mapping module.

The architecture of the system supports scalability and low operational cost. Cloud services (AWS, GCP, or Azure) ensure smooth handling of user traffic and model inference, while microservices architecture allows independent scaling of the image processing, risk analysis, and mapping modules. The use of open-source libraries keeps development cost-effective, and caching mechanisms like Redis reduce server load by reusing previous predictions.

By building an integrated pipeline from image input to real-time regional visualization, this research offers a holistic solution that bridges the gap between technological advancements and practical field-level usability. The system not only helps individual farmers detect crop diseases early but also benefits entire communities by enabling collective disease monitoring. Through accessible design, accurate predictions, and a strong focus on affordability, this work demonstrates how AI-driven agricultural tools can transform crop protection and support sustainable farming practices at scale.

6.3 Future Scope :

Expansion to Multiple Crops and Diseases:

- Extend the model to support more crop types and rare disease categories for broader applicability.

Offline Functionality:

- Build offline prediction features using lightweight on-device models for areas with limited internet access.

Advanced Forecasting Models:

- Integrate LSTM or time-series forecasting tools to better predict long-term disease outbreaks.

Farmer Advisory System:

- Add AI-driven suggestions for pesticide usage, preventive steps, and organic alternatives.

Government & Research Collaboration:

- Connect the framework to agricultural departments for large-scale disease surveillance and policy planning.

Improved Crowdsourcing Engagement:

- Use gamification, reward systems, and regional leaderboards to motivate more farmer participation.

IoT & Drone Integration:

- Incorporate drone imagery, soil sensors, and weather stations for higher accuracy and richer datasets.

Multilingual and Voice-Assisted Features:

- Provide support for local languages and voice-based navigation to increase adoption in rural communities.

6.5 Closing Remarks

This project represents a significant step toward modernizing agricultural disease monitoring through the integration of AI, crowdsourcing, and mobile technology. By empowering farmers with instant access to reliable disease predictions and regional risk insights, the system bridges the gap between traditional farming practices and modern digital tools. The framework is designed to be affordable, scalable, and adaptable, making it suitable for both small-scale farmers and large agricultural ecosystems. With continued development, collaboration, and widespread adoption, this solution has the potential to revolutionize crop health management and contribute meaningfully to global food security.

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